

COMICCRAFT

AI Comic Story Creator using Gemini Models

PROJECT PRESENTATION & DOCUMENTATION

Prompt	AI Story	Comic Panels	PDF Export
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A web-based application for personalized comic stories, dialogue, illustrations, preview, and PDF export.

1. Project Overview

ComicCraft is a web-based AI application that generates personalized comic book stories and illustrations from user-provided prompts. It uses FastAPI and integrates Google Gemini models with Stable Diffusion.

Users provide a story prompt, main character name, setting, tone, and art style. The system produces a panel-by-panel storyline with corresponding illustrations, allows preview, and supports complete comic PDF download.

2. User Workflow

1 User Input	2 Gemini Flash Outline	3 Gemini Pro Story & Dialogue	4 Stable Diffusion Images	5 Preview & PDF
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3. Technical Architecture

Frontend: HTML, CSS and Jinja2 templates for collecting inputs and displaying the comic preview.

Backend: FastAPI handles routing, input processing, AI workflow coordination, layout building and PDF export.

AI Integration: Gemini Flash creates structured panel outlines; Gemini Pro creates detailed narration and dialogue; Stable Diffusion creates comic-style illustrations.

4. Model Selection

Model / Tool	Purpose
Gemini Flash	Fast, structured five-panel comic outline generation
Gemini Pro	Detailed narration and character dialogue
Stable Diffusion	Comic-style illustration generation

5. Core Functionalities

generate_outline() — Creates a structured five-panel outline.

generate_story() — Expands the outline into narration and character dialogue.

generate_image() — Creates a comic-style image from an image prompt and saves it to the panels directory.

build_comic_layout() — Matches images with their corresponding panel stories.

save_pdf() — Compiles panel images and narration into a multi-page PDF.

6. FastAPI Routes

/ — Loads the homepage for entering story details.

/generate — Processes form input, generates comic panels and returns the preview.

/generate-comic/json — Accepts JSON input and returns layout data and generated PDF path.

/export-success — Displays confirmation after export.

/test-image — Allows direct testing of image generation.

7. Frontend & Deployment

The homepage collects story prompt, character name, setting, tone and art style. The comic preview displays each panel with title, illustration, scene description, captions and narration. The export-success page confirms comic creation and provides a path to create another comic.

The project uses a Python virtual environment and dependencies including FastAPI, Uvicorn, Google Generative AI, Hugging Face Diffusers and FPDF. Sensitive Gemini and Hugging Face credentials are stored as environment variables.

Local server: `uvicorn app.main:app --reload`

Application: `http://127.0.0.1:8000` **FastAPI docs:** `/docs`

8. Testing & Verification

The supplied project documentation describes verification of the complete workflow: AI text generation, image creation, layout building, frontend rendering and PDF export. Testing should verify valid inputs, route behavior, successful comic generation, image generation, preview rendering and successful PDF export. Invalid inputs and AI/API failures should be handled gracefully.

9. Conclusion

ComicCraft demonstrates an end-to-end AI-assisted comic creation workflow. Gemini Flash and Gemini Pro provide structured and creative storytelling, while Stable Diffusion produces comic-style illustrations. The FastAPI backend connects these services to an interactive frontend and PDF export pipeline. The modular architecture can support future enhancements such as user profiles, comic libraries and multi-page story arcs.

10. Project Technology Summary

Component	Technology / Purpose
Frontend	HTML, CSS, Jinja2 – user input and comic preview
Backend	FastAPI – routing and application logic
Story Outline	Gemini Flash – structured 5-panel outline
Story & Dialogue	Gemini Pro – narration and character dialogue
Images	Stable Diffusion + Hugging Face Diffusers
PDF Export	FPDF – multi-page comic PDF
Server	Uvicorn – ASGI development server
Language	Python

Reference basis: supplied ComicCraft project report and documentation materials. Content has been organized into a presentation-style PDF without adding unsupported project features.